GT2N: An Open-Source 2nm Nanosheet PDK Enabling Multi-Width/VT Benchmarking

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***Abstract*—This work introduces GT2N, an open-source, process-aware 2nm nanosheet PDK with wrap-around contact, integrated backside power delivery and buried power rail support. Leveraging 10 nanosheet width and 10 threshold voltage variants, a comprehensive DTCO study was conducted using full-chip benchmarks. Results demonstrate that a configuration using 2 specific W and 5 VT flavors achieves near-optimal PDP, within 3.7% of the full 10W×10VT baseline in the worst case, while significantly reducing library complexity. GT2N provides an accessible, realistic platform for 2nm DTCO exploration.**

***Index Terms*—2nm, PDK, Library, NSFET, WAC, BPR, DTCO**

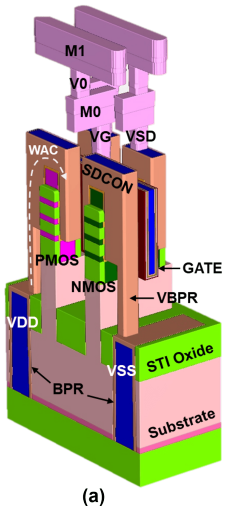
I. Introduction

Driven by demands from AI, HPC, and mobile applications, the semiconductor industry is scaling towards the 2nm node, adopting Gate-All-Around (GAA) nanosheet FETs (NSFETs) over FinFETs for better electrostatics, density, and continuous width scalability [1]. Simultaneously, Backside Power Delivery Networks (BSPDN) are being introduced to improve power integrity and reduce frontside routing congestion, with industry targeting mass production around late 2025 [2][3][4]. Accurate and accessible Process Design Kits (PDKs) are essential for evaluating and leveraging these advanced architectures.

However, while PDKs are crucial for design-technology co-optimization (DTCO), reliable PDKs for 2nm remain limited in academia. Existing academic PDKs often lack maturity or realistic standard cell libraries needed for system-level benchmarking, while proprietary industry PDKs are inaccessible to most researchers. To address this gap, we present GT2N, an open-source 2nm GAA NSFET PDK featuring Buried Power Rail (BPR) and BSPDN integration, available on GitHub [5]. Advanced 3D virtual fabrication tools [6][7] are used to faithfully model nanosheet cell geometries and interconnects.

The contributions are as follows: 1) To our knowledge, GT2N provides the most realistic open-source 2nm PDK available in academia, developed with foundry-representative specifications and validated using process-aware, virtual fabrication tools. 2) Leveraging GT2N, we conduct a comprehensive DTCO study analyzing the system-level performance impact of using multiple nanosheet widths (W) and threshold voltages (VT) based on various full-chip benchmark designs. 3) We offer optimized combinations of 2 W and 5 VT flavors within our 2nm PDK, along with recommendations for the best-performing W & VT subsets based on design constraints, giving valuable guidance for designers targeting the 2nm node.

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(a)

Layer	Width/ Pitch (nm)	R_{BEOL} (Ω/μm)	Process
M0/M2	12/24	622	0.33 NA EUV DPT
M1/M3	14/28	438	0.33 NA EUV DPT
M4/M5	21/42	166	193i SADP
M6-M9	38/76	26	193i SPT
M10/M11	56/112	7.5	193i SPT
M12/M13	360/720	0.64	248 SPT
BPR	32/144	28	193i SPT
BM1/BM2	56/112	7.5	193i SPT
BM3/BM4	360/720	0.64	248 SPT

(b)

Fig. 1. (a) 3D virtual fabrication of a GT2N logic cell illustrating BPR, nano-TSV (VBPR), WAC (SDCON), and connections to BEOL. (b) Summary table of BEOL layer dimensions, extracted resistances, and process assumptions.

II. GT2N PDK Development Framework

A. BEOL and MOL Architecture

The interconnect architecture is crucial for performance and power delivery at the 2nm node. GT2N incorporates a Back-End-of-Line (BEOL) and Middle-of-Line (MOL) stack designed to reflect industry trends, optimized for BSPDN integration. Fig. 1 (a) showcases key MOL and BEOL elements.

BPR is processed using Tungsten (W) fill in $4\,\mathrm{nm}$ TiN liners and oxide layers, targeting robust power delivery [8][9]. Power is routed from the BPR through a nano-TSV (Through-Silicon Via), comprising a $2\,\mathrm{nm}$ TiN liner with W fill [10][11], which connects to the side of the Source/Drain (S/D) Wrap-Around Contact (WAC) metal [12][13]. The WAC structure over the S/D epi greatly improves power delivery and is expected to be crucial for BSPDN [14][15]. The contact resistivity (ρ_c) at the silicide-S/D interface is $2.2 \times 10^{-9}\,\Omega\,\mathrm{cm}^2$ for n-Si and

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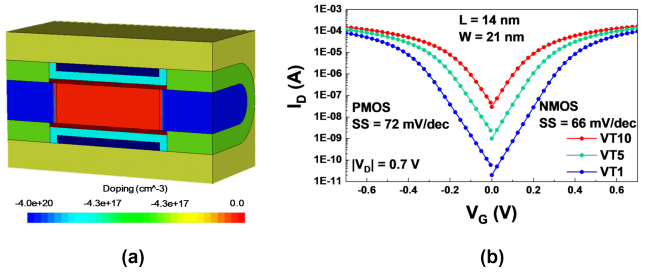


Fig. 2. (a) Cross-section of the TCAD 3D virtual fabrication model for a single nanosheet device. (b) SPICE models for 3-stack NMOS and PMOS devices, calibrated to I_d-V_g, I_d-V_d, C_g-V_g curves from TCAD simulations.

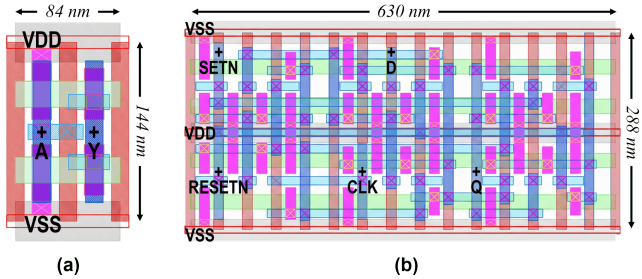


Fig. 3. 2D Layout examples from the GT2N library: (a) INVx1, (b) DFFx1.

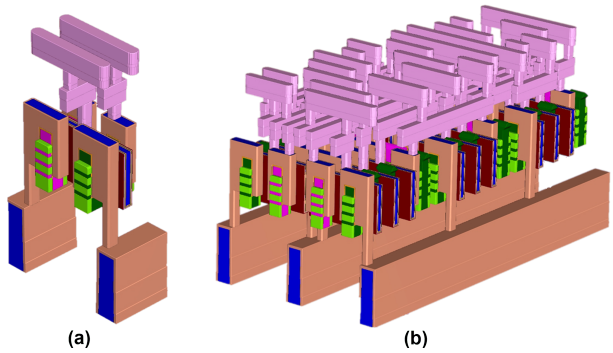


Fig. 4. Final 3D structures from virtual fabrication: (a) INVx1, (b) DFFx1.

TABLE I
CALIBRATED VT FLAVORS AND KEY METRICS FOR 3-STACK NSFET DEVICE OF W = 21 NM (CURRENTS IN A, V_{th} IN V)

VT Flavor	NMOS			PMOS		
	I_{off}	I_{on}	V_{th}	I_{off}	I_{on}	V_{th}
VT1	2.00e−11	9.48e−5	0.309	5.00e−11	8.09e−5	-0.309
VT2	5.00e−11	1.04e−4	0.284	1.20e−10	8.95e−5	-0.282
VT3	1.20e−10	1.12e−4	0.260	3.00e−10	9.86e−5	-0.255
VT4	4.00e−10	1.24e−4	0.226	8.00e−10	1.09e−4	-0.225
VT5	1.00e−9	1.33e−4	0.201	2.00e−9	1.18e−4	-0.198
VT6	2.50e−9	1.41e−4	0.176	5.00e−9	1.28e−4	-0.170
VT7	5.00e−9	1.48e−4	0.157	1.00e−8	1.35e−4	-0.149
VT8	1.00e−8	1.54e−4	0.138	2.00e−8	1.43e−4	-0.128
VT9	2.00e−8	1.60e−4	0.119	3.00e−8	1.47e−4	-0.116
VT10	3.00e−8	1.64e−4	0.108	4.00e−8	1.50e−4	-0.108

$2.0 \times 10^{-9}\,\Omega\,\mathrm{cm}^2$ for p-SiGe, consistent with measurements for advanced CMOS processes [12][16] and meeting the projected requirements for the targeted Contact Gate Pitch (CGP) [17]. The gate metal stack employs TiAlN for NMOS and TiN for PMOS as work function metals, deposited sequentially (TiAlN over TiN). This is followed by TaN and TiN layers, capped with W for the gate contact material [18][19].

Ruthenium (Ru) is used for the lower BEOL layers (M0-M3) as well as for the upper MOL (VG, VSD), leveraging its merits in scaled dimensions [7][20]. For higher metal levels (M4~), Copper (Cu) wires are employed, featuring a TaN/Ru barrier/liner configuration, optimized based on our previous work [21][22]. Key parameters are summarized in Fig. 1 (b).

B. Nanosheet Device Design and Modeling

For device, GT2N employs a hybrid approach: the core device physics of a single nanosheet including the channel, gate oxide stack, and initial gate metal deposition, are simulated using 3D TCAD, while large-scale virtual fabrication tools handle the complex 3D geometries of S/D epitaxy, contacts, and inter-channel effects within the full standard cell context.

Each device consists of three vertically stacked nanosheets. Key device dimensions include nominal gate length of 14 nm, channel thickness of 5 nm, vertical spacing between nanosheets of 12 nm, and a lateral N-PMOS separation of 30 nm [23][24][25]. The gate dielectric is designed for an Effective Oxide Thickness (EOT) of 7 Å [26]. A non-selective gate process flow is assumed for cost-effectiveness, resulting in a dual-spacer comprising a thick low-k dielectric capped with a thin high-k dielectric, totaling 6 nm in lateral thickness [27]. Fig. 2 (a) shows the 3D TCAD design of a single nanosheet.

Fig. 2 (b) displays the calibrated I_d-V_g curves for NMOS and PMOS devices with the nominal nanosheet width (W) of 21 nm. S/D doping and gate metal work function engineering yield 10 distinct threshold voltage (VT) flavors for both NMOS and PMOS, spanning a total range of roughly 200 mV [1][4]. The I_{on}/I_{off} characteristics are calibrated to align with reported data from leading foundries for advanced nanosheet nodes, targeting a minimum off-current (I_{off}) around 20-30 pA, a PMOS I_{off} roughly double that of NMOS, and a PMOS I_{on} reaching approximately 90% of the NMOS I_{on}. Table I lists the calibrated parameters for the resulting 10 VT flavors.

To enable comprehensive DTCO studies involving device sizing, we extend the device modeling to include 10 discrete W, ranging from 13 to 31 nm (max. within the 6-track cell height). This results in a suite of 100 distinct device models.

C. Logic Cell Design and Library Generation

The standard cell library serves as the foundation for digital design implementation. We developed logic layouts using design rule decks derived and scaled from our prior work [7].

The CGP is set to 42 nm. A 6-track (6T) standard cell architecture with a cell height of 144 nm is employed [14][23][24], providing sufficient routing tracks within the cell and accommodating the wide range of nanosheet widths (13-31 nm) required for our DTCO study. Fig. 3 (a) shows the 2D layout of an INVx1 cell as an example of a single-height cell, while Fig.

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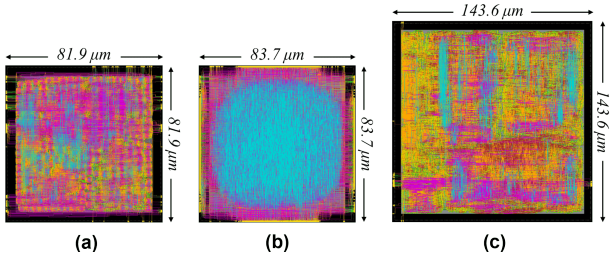


Fig. 5. Example 2D layouts of benchmark designs implemented using the GT2N PDK: (a) AES-128, (b) LDPC, (c) JPEG.

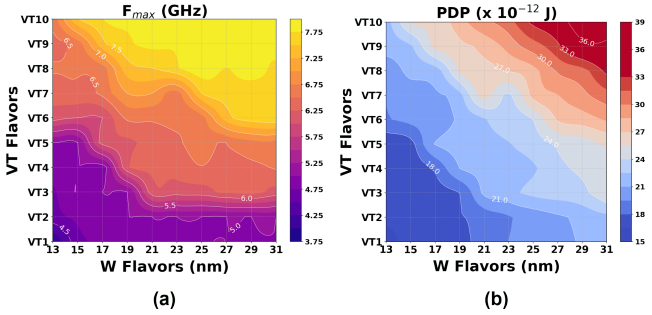


Fig. 6. System-level evaluation of AES-128 using single flavors: (a) Maximum Frequency (F_{max}, GHz), (b) Power-Delay Product (PDP, $\times 10^{-12}$ J). These results highlight the PPA trade-offs and the need for multi-flavor DTCO.

3 (b) shows the layout for an asynchronous DFFx1, illustrating a double-height cell design used for more complex functions.

These 2D layouts serve as blueprints for constructing detailed 3D models of each logic cell using virtual fabrication tools. The tool emulates a foundry-grade process flow based on the geometrical specifications defined in previous Sections, and accurately captures complex 3D structures, including nanosheet stacks, WACs, BPR connections, and multi-level interconnects. Next, rigorous parasitic analysis is performed on these 3D models using field solver tools. This methodology yields highly realistic parasitic netlists, often containing tens of thousands of parasitic elements per cell, offering a significant fidelity advantage over conventional 2D layout extraction methods that rely on simplified lookup tables (ITF, ICT files). Fig. 4 (a) and (b) depict the final 3D structures resulting from virtual fabrication of the INVx1 and DFFx1 cells, respectively.

The complete library comprises 71 unique standard cells, including basic gates, complex combinational cells, and sequential elements. Critically, to support the W and VT DTCO study, we generated 10 distinct sets of layouts, one for each W flavor. Each set was then processed through the 3D virtual fabrication and parasitic extraction flow, resulting in 10 unique sets of parasitic netlists (71 cells each). By combining these width-specific parasitic netlists with the 100 W/VT-specific device models, we generated a total of 100 distinct logic libraries (LIB/LEF pairs). Each library represents a specific combination of one W flavor and one VT flavor, enabling a thorough exploration of the design space within the 2nm node.

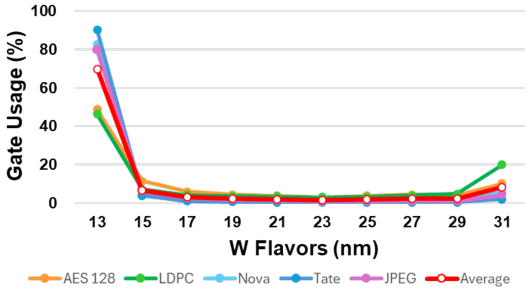


Fig. 7. Gate cell usage percentage per W flavor when implementing designs using all 10 W and 10 VT (100 total) libraries.

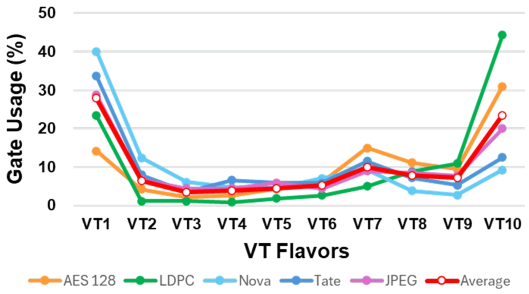


Fig. 8. Gate cell usage percentage per VT flavor when implementing designs using 2 W (13 nm and 31 nm) and 10 VT (20 total) libraries.

III. Full-Chip Implementation

We utilized the comprehensive set of 100 generated libraries to perform full-chip RTL-to-GDSII design implementations and conduct system-level DTCO studies, utilizing 5 industrial benchmark circuits. Fig. 5 shows example layouts at 2nm.

A. Single Library Evaluation

We first evaluated the system-level performance of each of the 100 libraries individually. The AES-128 benchmark was synthesized and placed-and-routed (P&R) for maximum frequency (F_{max}). As shown in Fig. 6 (a), F_{max} increases with wider W and stronger VT flavors but eventually plateaus at the high-W, strong-VT end (yellow region), where power consumption also rises sharply. Fig. 6 (b) plots the Power-Delay Product (PDP), showing that while these libraries achieve the highest speed, their power efficiency degrades. Conversely, low-power libraries sacrifice performance. Thus, identifying a single optimal W/VT point is non-trivial and will depend on specific design constraints. This demonstrates the complex interplay of W and VT trade-offs, highlighting the need for strategic DTCO using multiple library flavors for optimal PPA.

B. Nanosheet Width (W) Flavor Optimization

We next explored optimizing the number of W flavors, assuming all 10 VT flavors are available. Baseline designs were created using all 100 libraries (10Wx10VT). During synthesis and P&R under tight timing constraints, cells from different libraries are strategically selected to meet PPA targets.

Fig. 7 shows the standard-cell usage distribution across W flavors for all five benchmarks. The narrowest width 13

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TABLE II
PERFORMANCE METRICS VS BASELINE (10WX10VT) FOR REDUCED W FLAVOR SETS (USING 10 VT FLAVORS)

Config.	AES-128 (clk: 150 ps)			LDPC (clk: 300 ps)			Nova (clk: 500 ps)			Tate (clk: 200 ps)			JPEG (clk: 200 ps)		
	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)
10Wx10VT	6.73	108.6	100.0	3.25	58.5	100.0	2.00	48.4	100.0	5.03	196.8	100.0	4.98	308.2	100.0
4Wx10VT	6.65	108.6	101.4	3.21	58.8	102.0	2.00	47.2	97.6	4.97	197.6	101.6	4.99	310.3	100.5
3Wx10VT	6.64	108.3	101.2	3.25	58.2	99.6	2.00	48.4	100.0	4.99	197.4	101.1	4.99	316.0	102.2
2Wx10VT (i)	6.66	107.2	100.0	3.22	59.5	102.9	2.00	48.6	100.5	4.99	195.2	100.0	4.98	314.5	102.1
2Wx10VT (ii)	6.49	109.1	104.3	2.94	61.3	115.9	1.79	48.8	112.9	4.99	198.9	101.9	4.92	319.0	104.8
1Wx10VT	6.23	110.9	110.4	2.89	60.7	116.7	1.66	48.7	121.1	4.99	196.2	100.5	4.83	321.5	107.5

TABLE III
PERFORMANCE METRICS VS BASELINE (10WX10VT) FOR REDUCED VT FLAVOR SETS (USING 2 W FLAVORS)

Config.	AES-128 (clk: 150 ps)			LDPC (clk: 300 ps)			Nova (clk: 500 ps)			Tate (clk: 200 ps)			JPEG (clk: 200 ps)		
	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)	F_{eff} (GHz)	P (mW)	PDP (norm.)
10Wx10VT	6.73	108.6	100.0	3.25	58.5	100.0	2.00	48.4	100.0	5.03	196.8	100.0	4.98	308.2	100.0
2Wx5VT (i)	6.65	108.8	101.6	3.28	57.9	98.4	2.00	48.3	99.9	5.00	196.6	100.7	5.00	327.1	105.7
2Wx5VT (ii)	6.65	111.2	103.7	3.28	59.6	101.1	2.00	48.6	100.5	4.99	200.7	102.8	5.00	318.6	103.0
2Wx4VT (i)	6.55	110.8	105.0	3.25	60.6	103.8	2.00	47.9	98.9	4.99	199.6	102.3	4.98	324.6	105.2
2Wx4VT (ii)	6.60	112.7	105.9	3.29	62.2	105.2	2.00	48.4	100.0	4.99	202.9	103.9	4.99	328.1	106.2
2Wx3VT	6.66	111.6	104.1	3.21	62.8	108.9	2.00	48.4	100.1	5.00	202.9	103.9	4.88	333.4	110.4
2Wx2VT	6.65	115.6	107.9	3.22	63.1	108.9	2.00	49.1	101.5	4.99	209.3	107.3	4.85	333.7	111.2

nm dominates due to its low capacitance and power in non-critical paths, while 15 and 31 nm are next most used for timing-critical paths; intermediate widths are rarely selected. Based on this usage priority, we evaluated scenarios using reduced sets of W flavors (while keeping all 10 VT flavors): 4W (13, 15, 17, 31 nm), 3W (13, 15, 31 nm), 2W case (i) (13, 31 nm), 2W case (ii) (13, 15 nm), and 1W (13 nm). Table II compares the resulting PPA relative to the 10W baseline.

While PDP generally degrades with fewer W flavors, the 2W case (i) using 13 and 31 nm maintains PDP close to the 10W baseline across all designs. In contrast, 2W case (ii) results in significant PDP degradation for certain designs (up to 15.9% for LDPC), and the 1W case degrades by up to 21.1% (Nova). Given that node-to-node power improvements are in the 25-35% range [1], a >20% loss from library restriction is unacceptable. Thus, using two W flavors (13, 31 nm) offers the best balance between process complexity and performance.

C. Threshold Voltage (VT) Flavor Optimization

Having established the optimal 2W flavor set, we next optimized the number of VT flavors. Designs were implemented using the 2W flavors combined with all 10 VTs, with Fig. 8 showing cell usage per VT flavor. Usage peaks occur at the highest (VT1, low leakage), mid (VT7), and lowest (VT10, high speed) flavors, while the fourth priority is less clear. Based on usage trends and the need for balanced granularity, we assessed several 5, 4, 3, and 2VT combinations. The best performing sets, relative to the 10W×10VT baseline, are summarized in Table III: 5VT(i) (VT1, 3, 5, 7, 10 – uniform steps), 5VT (ii) (VT1, 4, 7, 8, 10 – usage-informed), 4VT (i) (VT1, 4, 7, 10 – uniform steps), 4VT (ii) (VT1, 7, 8, 10 – usage-informed), 3VT (VT1, 7, 10), and 2VT (VT1, 10).

Results show that using only 2VTs severly degrades PDP (up to 11.2% for JPEG) for most designs, while 3VTs harm some designs (10.4% for JPEG). Both 4VT cases maintain near-baseline performance comparable to each other, and among 5VT cases, case (i) marginally outperforms case (ii), but within 2.1% difference. Based on this DTCO study, the proposed GT2N configuration adopts 2 W flavors (13, 31 nm) and 5 VT flavors (HVT, SVT, LVT, ULVT, ELVT, each representing VT1, 4, 7, 8, 10). This selection captures the most utilized device flavors within a foundry-representative range with reasonable granularity, achieving PDP results within 3.7% of the 10W×10VT baseline while significantly reducing library complexity and enabling efficient multi-W/VT optimization.

IV. CONCLUSION

We present GT2N, an open-source, process-aware PDK for 2nm nanosheet nodes that bridges the gap between academic and industry scaling trends. By integrating advanced features like WAC and BPR/BSPDN, and employing rigorous 3D virtual fabrication, GT2N offers a realistic platform for system-level evaluation. Our DTCO study across various benchmarks reveals a combination using two nanosheet widths (13, 31 nm) with five threshold voltages (VT1, 4, 7, 8, 10) provides an optimal balance of PPA (within 3.7% PDP of the ideal baseline) and design complexity. This open-source platform enables further research and benchmarking of emerging circuit designs and DTCO methodologies in the 2nm nanosheet era.

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